

Analyzing Ecommerce data in Tableau

Session 4

Dataset

- The data is for an Online Store selling pet supplies.
- Please read metadata file for data description before start working
- Have a look on each data file

E-commerce operations and terminologies



Can pet food (4)

Illustrative total costs for canned pet food through various stages in the supply chain.



Factory



Freight



Warehouse



Shipping



Profit/loss



Price

$$\$4 + \$1.5 + \$1.5 + \$5 + 8 = \$20$$

COGS
(cost of goods sold)

Landed
Cost (freight)

Fulfillment Fees

Shipping
Cost

Profit
(Loss)

Retail
Price

Business Goals

- **Increase Sales**
- **Reduce Expenses**
- For sales growth, you will focus on upsell and cross-sell opportunities.
- In a cross-sell, you promote a relevant product at the point of purchase. For example, when a customer is ready to buy a beach umbrella, present an offer on beach chairs.
- In an upsell, you promote a higher-priced alternative or a higher quantity of a product. For example, when a customer buys pet food regularly, present an offer on organic food.

Business Goals

- Increase Sales
- Reduce Expenses

Shipping expense reduction strategies

For shipping cost reductions, we will explore the following alternatives. Consolidating multiple shipments into a single one. Shipping from a location closer to customers. Reducing package size dimensions and weight. And, finally, shipping a higher quantity of a product.

Business Goals

- Increase Sales
- Reduce Expenses

Market basket analysis

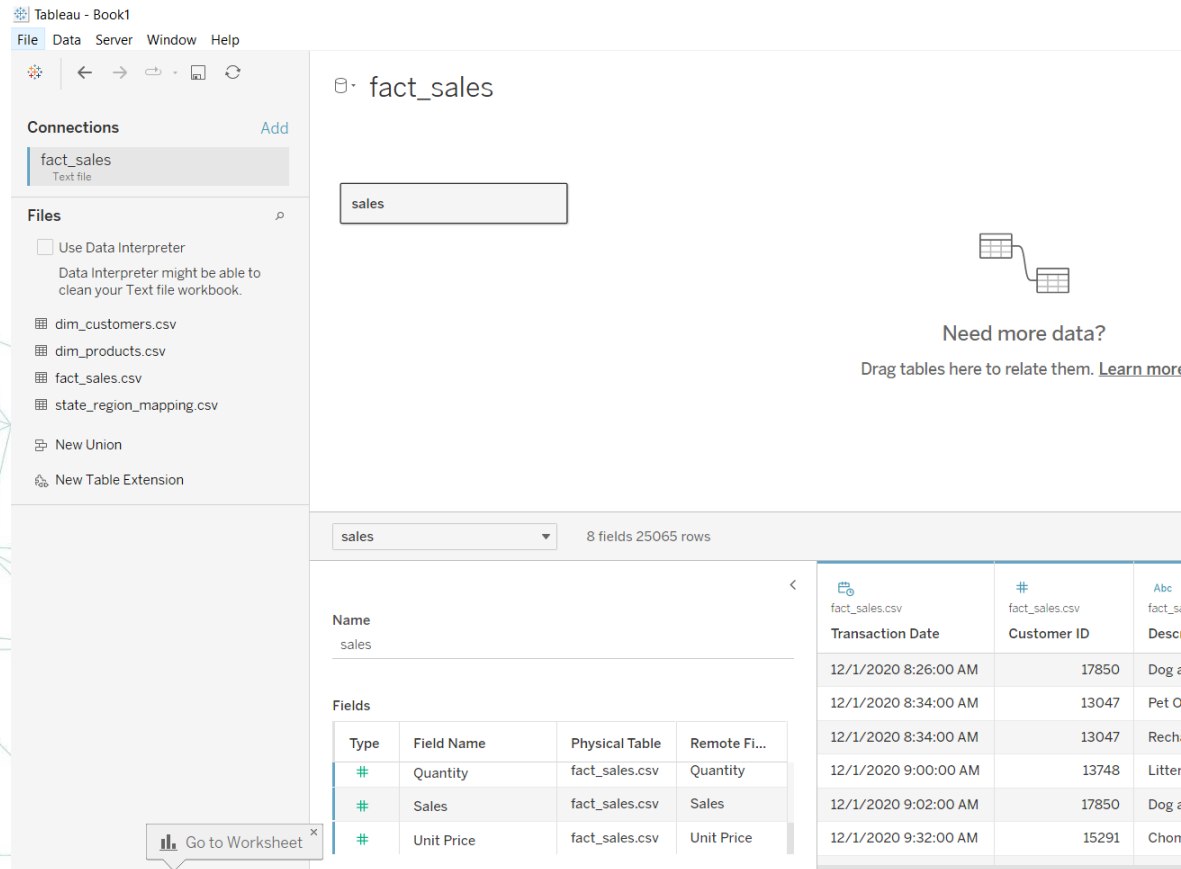
A market basket analysis aims to find products frequently bought together in a single checkout. This is determined by finding out the correlation coefficient between purchased quantities.

The value indicates the strength of the linear relationship. In this illustrated example, milk and bread correlate more strongly than milk and butter.

*Let's have fun and
start the real work*

Step 1 – Data source connection

- Connect the fact sales table and change its name to “sales”

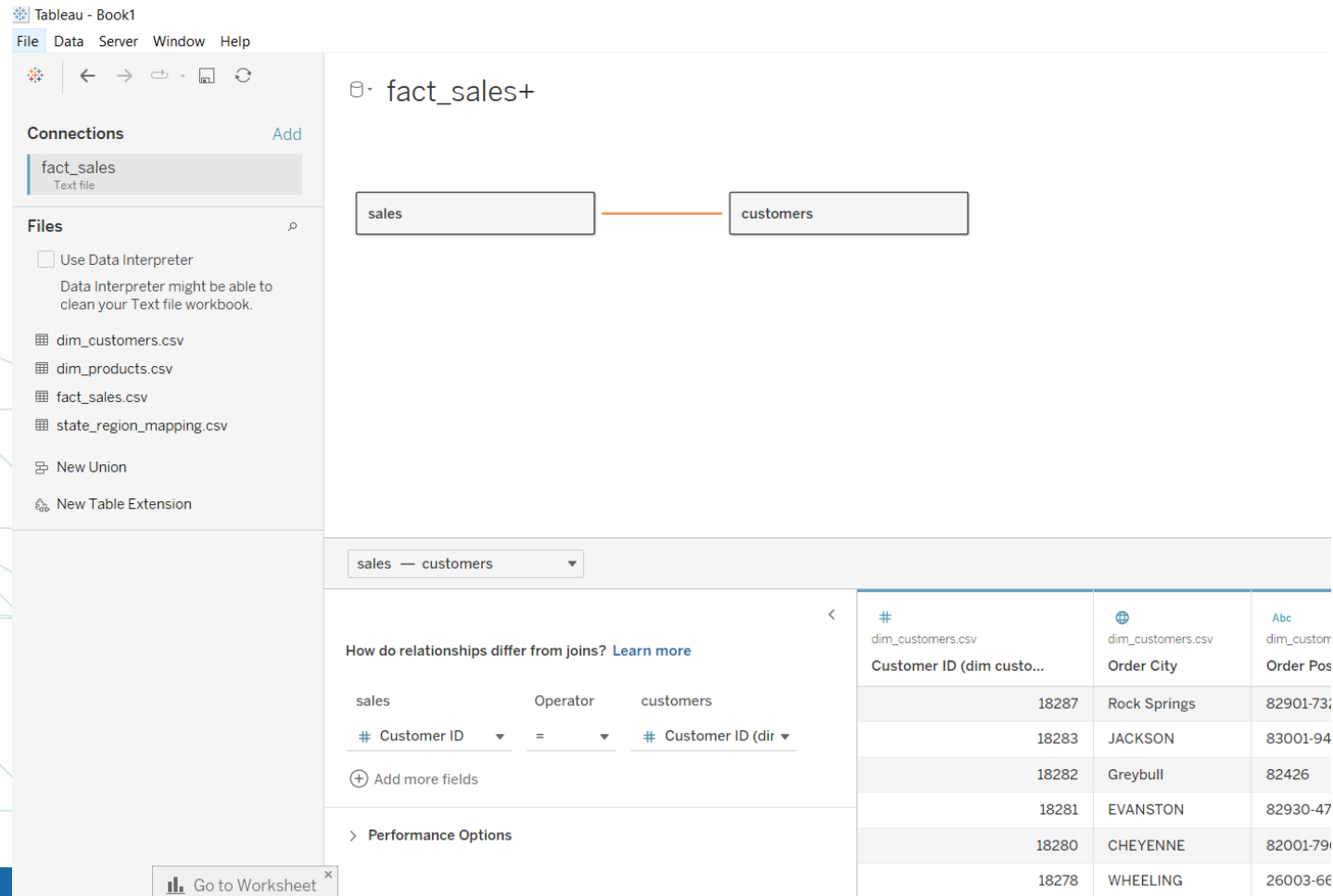


The screenshot shows the Tableau Desktop interface. In the 'Connections' pane on the left, 'fact_sales' is listed as a 'Text file'. Below it, under 'Files', several CSV files are listed: 'dim_customers.csv', 'dim_products.csv', 'fact_sales.csv', and 'state_region_mapping.csv'. The main workspace shows a box labeled 'fact_sales' being renamed to 'sales'. Below this, a message asks 'Need more data?' and suggests dragging tables to relate them. At the bottom, a table preview for the 'sales' table is shown, displaying columns: Transaction Date, Customer ID, and Description. A 'Go to Worksheet' button is visible at the bottom left of the preview.

Type	Field Name	Physical Table	Remote Fi...
#	Quantity	fact_sales.csv	Quantity
#	Sales	fact_sales.csv	Sales
#	Unit Price	fact_sales.csv	Unit Price

Step 2 – Data source connection

- Drag the dim_customers table and change its name to “customers” and make relationship with the sales table



The screenshot shows the Tableau Desktop interface with the following components:

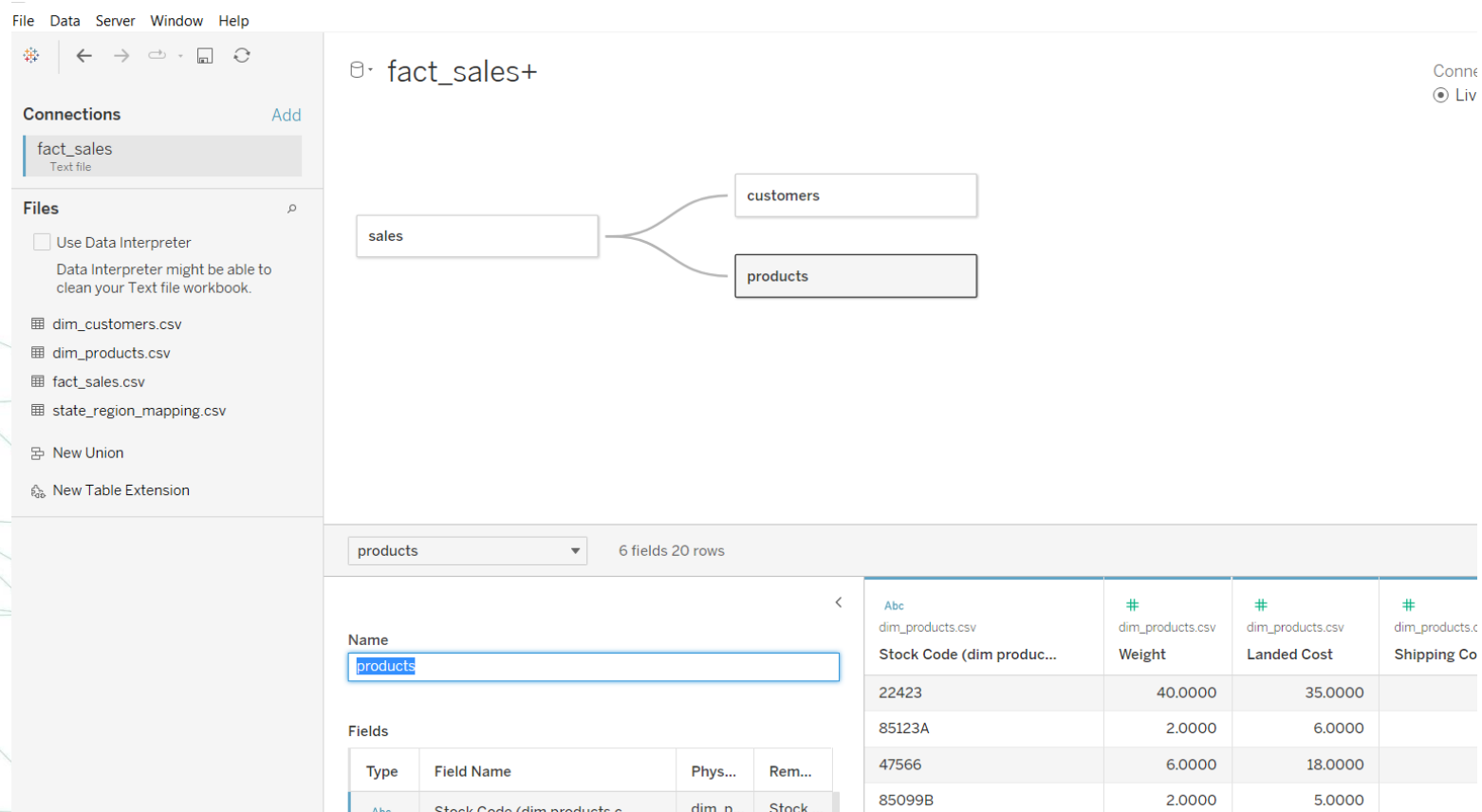
- Connections Panel:** Lists 'fact_sales' as a Text file.
- Files Panel:** Lists 'dim_customers.csv', 'dim_products.csv', 'fact_sales.csv', and 'state_region_mapping.csv'. It also includes options for 'Use Data Interpreter', 'New Union', and 'New Table Extension'.
- Fact Sheet View:** Displays a relationship diagram between 'sales' and 'customers' tables, connected by an orange line.
- Relationships Panel:** Shows the relationship between 'sales' and 'customers' based on the 'Customer ID' field. It includes a dropdown menu for 'sales - customers' and a table of data.
- Performance Options:** A section for optimizing the relationship.

Relationships Panel Data Table:

#	dim_customers.csv	dim_customers.csv	dim_customers.csv
	Customer ID (dim custo...	Order City	Order Pos
	18287	Rock Springs	82901-73
	18283	JACKSON	83001-94
	18282	Greybull	82426
	18281	EVANSTON	82930-47
	18280	CHEYENNE	82001-79
	18278	WHEELING	26003-66

Step 3 – Data source connection

- Drag the dim_products table and change its name to “products” and make relationship with the sales table.



fact_sales+

connections

fact_sales
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

- dim_customers.csv
- dim_products.csv
- fact_sales.csv
- state_region_mapping.csv
- New Union
- New Table Extension

products 6 fields 20 rows

Name
products

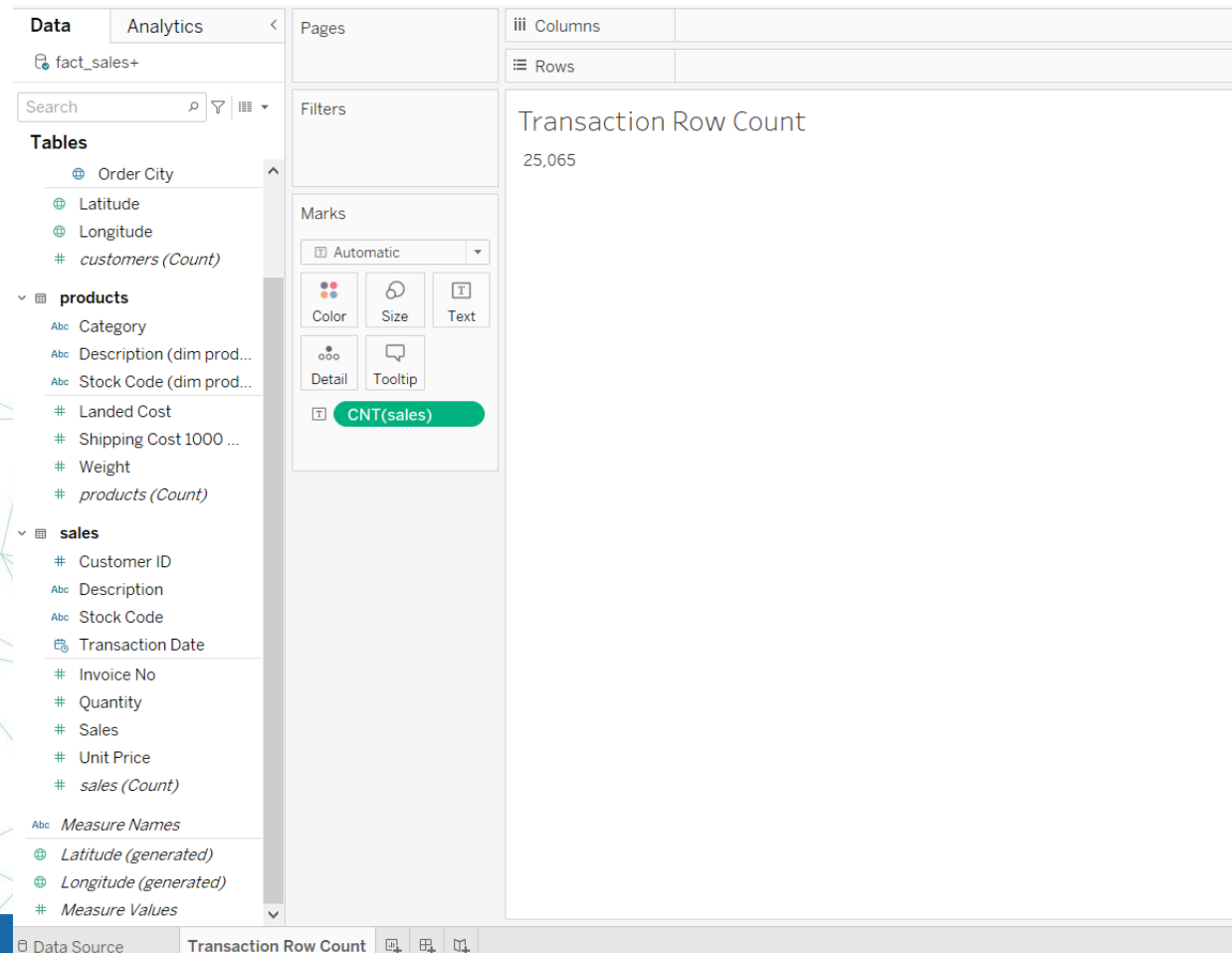
Fields

Type	Field Name	Phys...	Rem...
ABC	Stock Code (dim products)	dim p...	Stock ...

dim_products.csv	dim_products.csv	dim_products.csv	dim_products.c
Stock Code (dim produc...	Weight	Landed Cost	Shipping Co
22423	40.0000	35.0000	
85123A	2.0000	6.0000	
47566	6.0000	18.0000	
85099B	2.0000	5.0000	

Step 4 – Row count

- Make your first worksheet name it “Transaction Row Count” and check the row count



Step 5 – Data Cleaning

- All Valid Transactions must have a valid Invoice No.
- Create a new calculated field to flag the invoice no when it is null.

NULL Invoices

```
IF ISNULL([Invoice No]) THEN "NULL" ELSE "Valid" END
```

- Remove any records where the invoice no is null using data source filter.
- How many records in the sales table after removing nulls? It should be 24404

Columns	
Rows	
Transaction Row Count	24,404

fact_sales+

sales

Connection
☒ Live ☐ Extract

Filters
0 | Add

Filter [NULL Invoices]

General Wildcard Condition Top

☒ Select from list ☐ Custom value list ☐ Use all

Enter search text

☐ NULL
☒ Valid

All None Exclude

Summary

Field: [NULL Invoices]

Selection: Selected 1 of 2 values

Wildcard: All

Condition: None

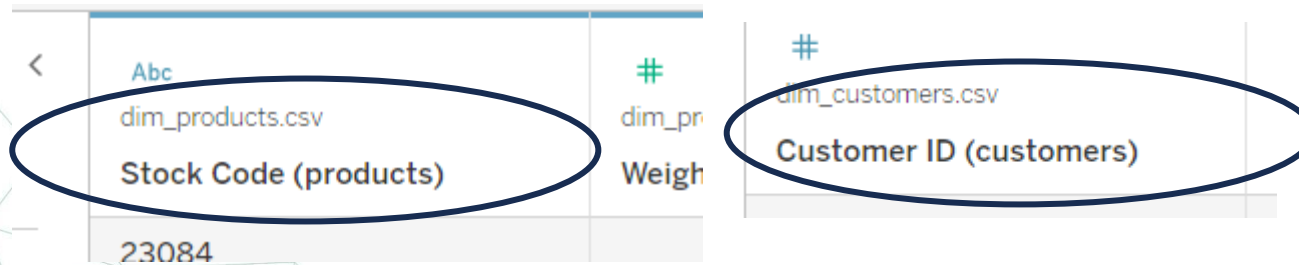
Limit: None

Reset OK Cancel

Type	Field Name	fact_sales.csv	fact_sales.csv	fact_sales.csv	fact_sales.csv
		Description	Stock Code	Invoice No	Quantity
		850 Dog and Puppy Pads	85123A	536,365	1
		047 Pet Odor Eliminator	84879	536,367	6
		047 Rechargeable Pet Nail Grinder	22960	536,368	1
		3748 Litter Slide Multi-Cat Scente...	22086	536,371	14

Step 6 – Data Cleaning

- The data source has three tables that were renamed. The dimensions and measures are still suffixed with a table name.
- Rename each field with the correct table name suffix.
- For example, rename **Stock Code (data_products.csv)** to **Stock Code (products)**.



Abc	#
dim_products.csv	dim_customers.csv
Stock Code (products)	Customer ID (customers)
23084	

Step 7 – Building Metrics

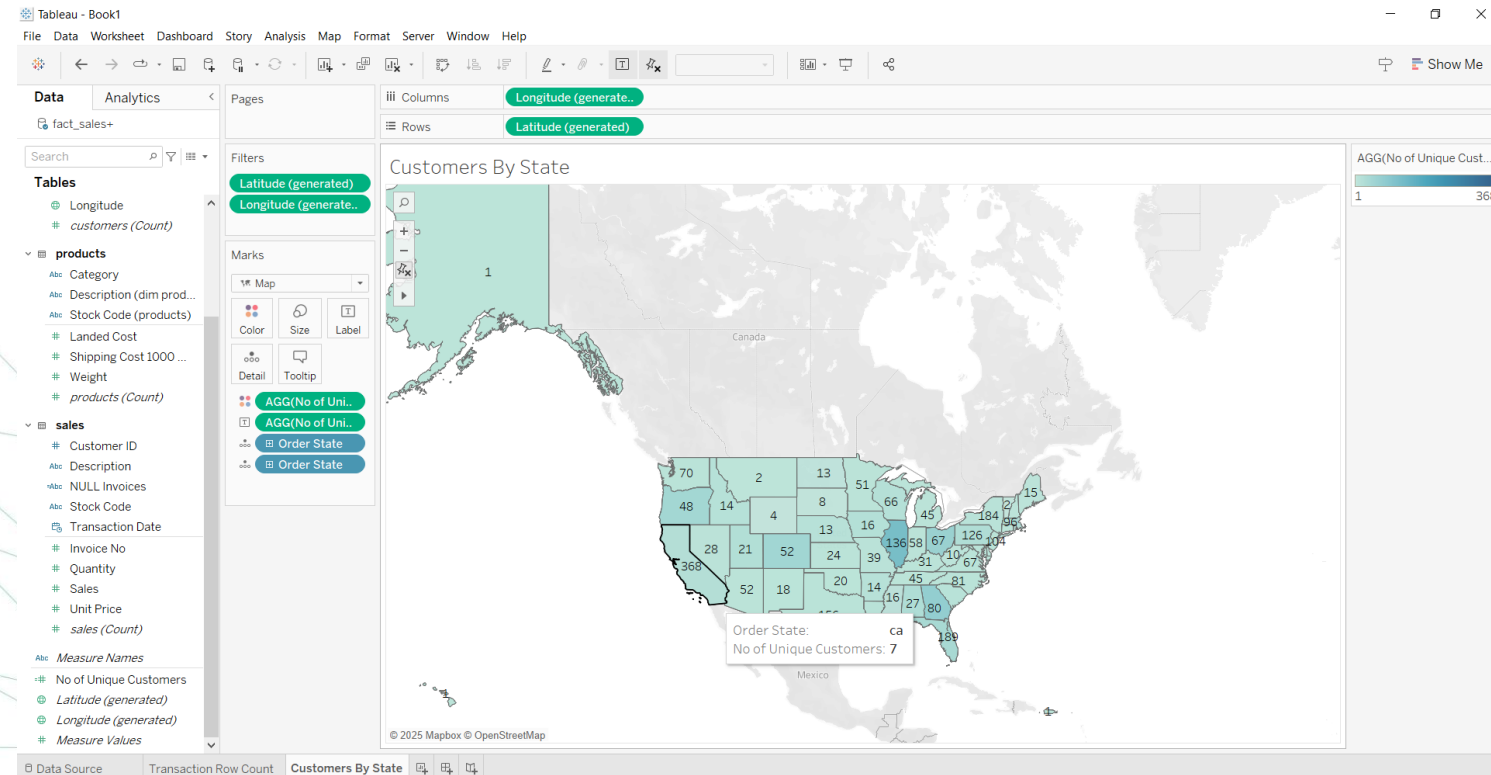
- Now that you have a clean dataset to work with let us build a few metrics for customers to understand how many customers we have and their total business.
- The metrics you will create are: ***Number of Customers*** and ***Customer Lifetime Value***. You will then display these metrics on a **map** and **box-and-whisker chart**, respectively.
- **Create a calculated field name it “No of Unique Customers”** to count how many unique customers we have.

No of Unique Customers

```
COUNTD([Customer ID])
```

Step 8 – Building Metrics

- Create a new sheet, "Customers by State", add a map, and display Number of Customers by state. Display the customer count as a text label on the state map. If you see Unknown locations (e.g., AE), filter those out. These represent overseas US military locations.



Step 9 – Building Metrics

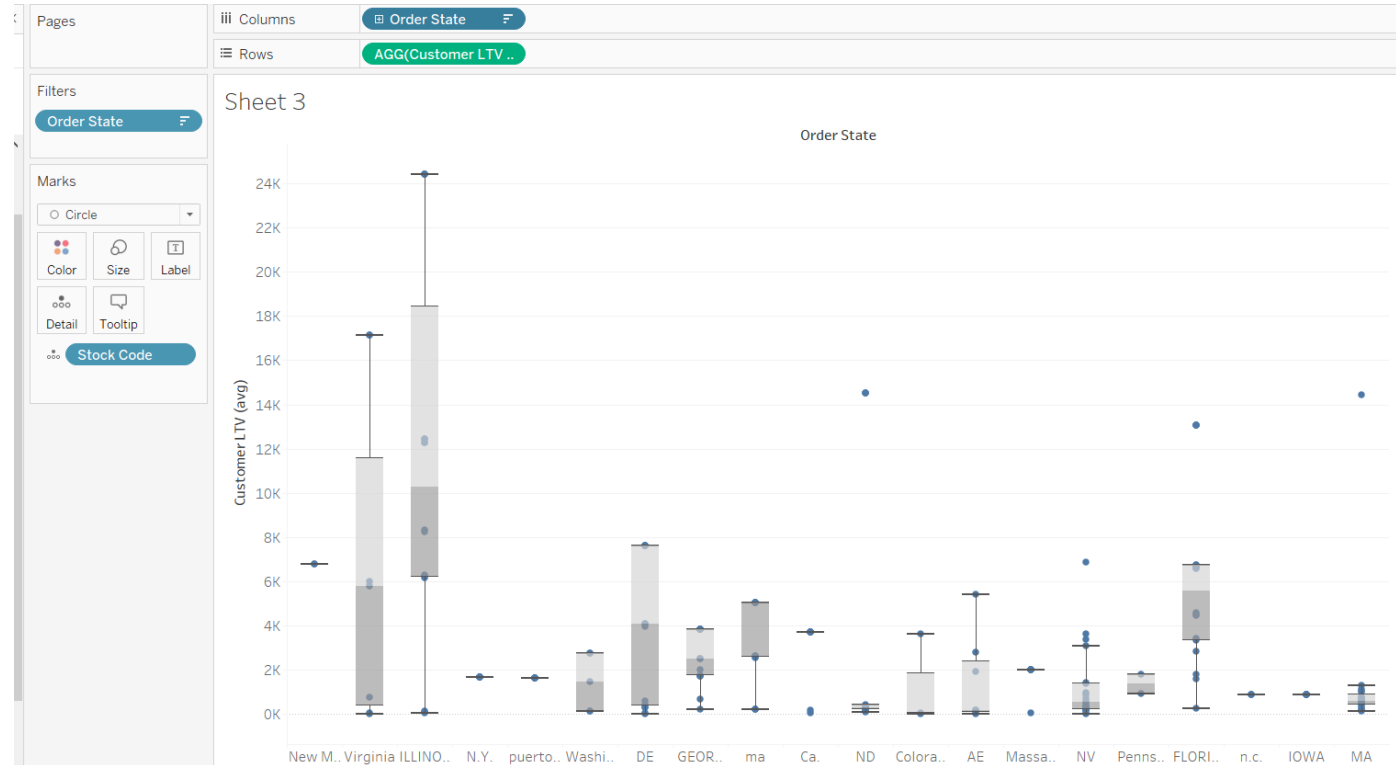
- Lifetime Value (LTV) means the total value of all sales by a customer.
- Create a field for "Customer LTV (avg)" and define it as the sum of all sales per customer. Use level of detail expression to calculate the value for each customer.

Customer LTV (avg)

```
AVG({FIXED [Customer ID] : SUM([Sales]) })|
```

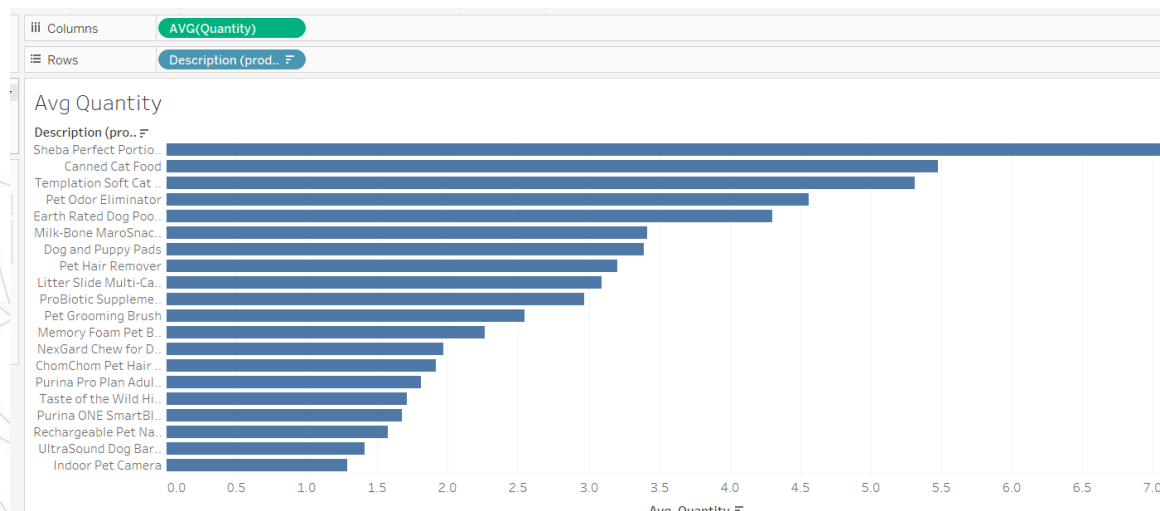
Step 10 – Building Metrics

- Create a new sheet "LTV by Order State" and add a text table of Order State, Stock Code (from sales) , and Customer LTV.
- Convert the text table into a box and whisker plot of lifetime value (LTV) by Order State.
- Keep the top 20 states by LTV and sort descending on LTV.
- **Hoe much is the median for Virginia?**
- **If it is 5,772 you are on the right track , please proceed.**



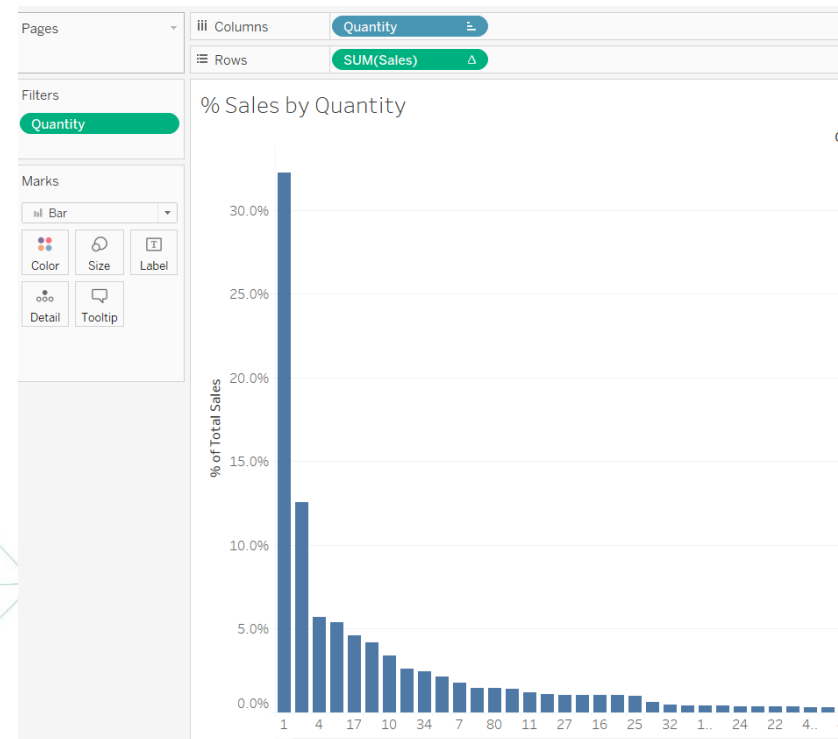
Step 11 – Building Metrics

- Let us view the total sales by average quantity.
- Many customers buy a single quantity of a product but combine that with other product types.
- The average quantity differs when calculated at the invoice level vs. across the entire business.
- This step will solidify your understanding of the relationship between the shipped quantity and the shipping cost.
- **Create a new sheet "Average Quantity" and add a bar chart of average quantity by Description (products).**
- **Sort descending on average quantity.**



Step 12 – Building Metrics

- You want to understand the distribution of sales by quantity of a product across all sales.
- Create a new sheet, "% Sales by Quantity", and add a bar chart of the percentage of total sales by quantity.
- Round the sales percent of the total calculation to two decimals and filter out quantities less than zero. These represent returned orders.
- Note you need to convert quantity to dimension on the shelf and the axis to discrete.**

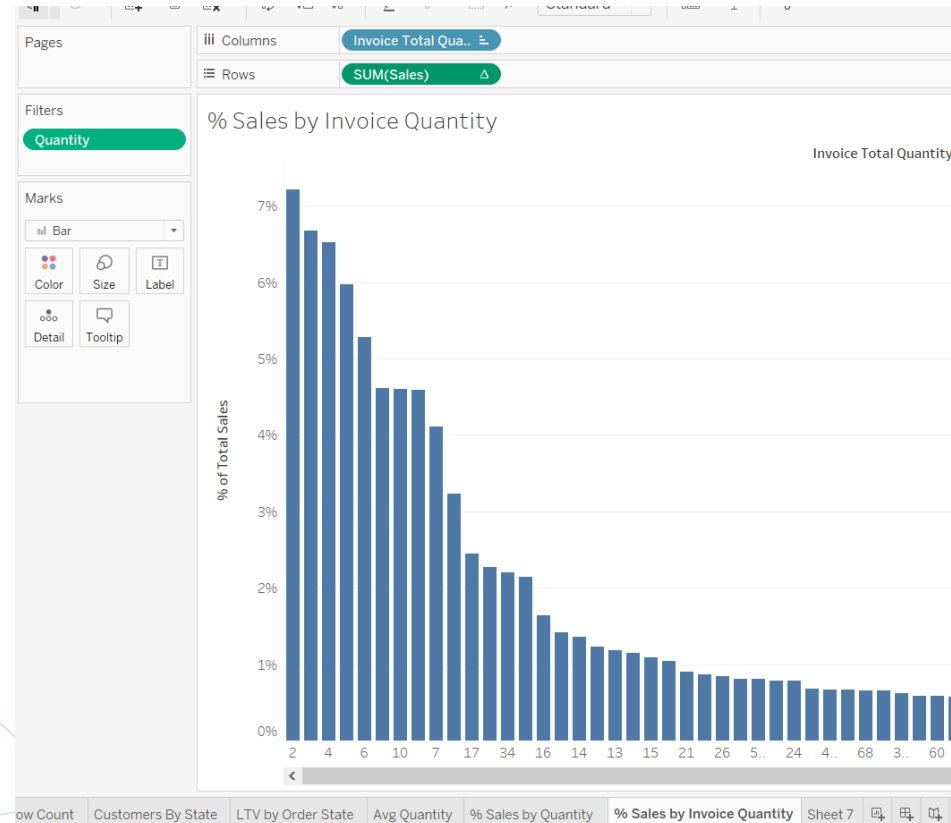


Step 13– Building Metrics

- Duplicate the sheet from the prior step and name it "% Sales by Invoice Quantity".
- create a new calculated field, "Invoice Total Quantity".
- Define it as the sum of Quantity for each Invoice No.
- Make a note of the percentage of total sales where quantity equals one.
- **If your answer is 5.98% You are on the right track please proceed to the next step.**

Invoice Total Quantity

```
{FIXED [Invoice No] : SUM([Quantity])}
```



Step 14– Building Metrics

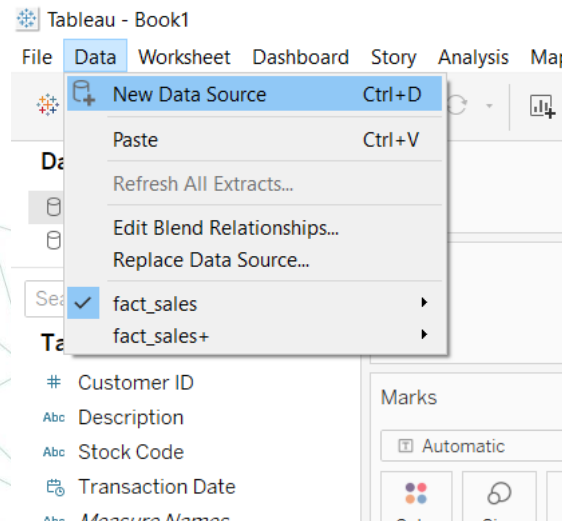
- The Shipping Cost 1000 mile field is the average cost to ship a single quantity of a product by itself (without combining it with other products).
- Create a new worksheet "per Unit Shipping Costs" and add a bar chart of **the average Shipping Cost 1000 mile by Description (products)**.
- Sort descending on average Shipping Cost 1000 mile.

Step 14– Building Metrics

- The Shipping Cost 1000 mile field is the average cost to ship a single quantity of a product by itself (without combining it with other products).
- Create a new worksheet "per Unit Shipping Costs" and add a bar chart of **the average Shipping Cost 1000 mile by Description (products)**.
- Sort descending on average Shipping Cost 1000 mile.

Step 15 – Market Basket Analysis

- The management wants to understand which products are frequently bought together.
- To find this out, you will build a correlation between the purchased quantity of products.
- You will connect with a transactions data CSV file and then connect the transactions table with itself (self-join).
- A self-join (on Invoice No) helps us understand products bought together as part of the same checkout.
- Create a **new** data source, "**Market Basket**", based on the CSV file **fact_sales.csv**, which you'll use for the rest of this exercise. Feel free to rename the table with a user-friendly business name. For example, **sales**.



Step 16 – Market Basket Analysis

- Add the sales table (for a second time) and rename it "sales-self-join".
- Join the two tables. The first join criteria is on Invoice No, and the second is on Stock Code.
- Change the join criteria on Stock Code to the less than operator to reduce redundant stock code combinations. For example, in { (A,B), (B,A) }, we can eliminate one of the pairs.

How do relationships differ from joins? [Learn more](#)

sales	Operator	sales Self join
# Invoice No ▼	= ▼	# Invoice No (fact ▼
Abc Stock Code ▼	< ▼	Abc Stock Code (fac ▼

Step 17 – Market Basket Analysis

- Define a calculated field Null Invoices to identify if there are records with no invoices in the sales or sales-self-join table.
- Define Stock Code Filter when Stock Code from either table has the same value.
- This represents exactly the same record in either of the transaction table.

Nulls

Market Basket

```
ISNULL([Invoice No]) OR ISNULL([Invoice No (fact sales.csv1)])
```

- Define **Stock Code Filter** when **Stock Code** from either table has the same value. This represents exactly the same record in either of the transaction table.

Stock Code Filter

Market Basket

```
[Stock Code] <> [Stock Code (fact sales.csv1)]
```

Step 18 – Market Basket Analysis

- Create a new sheet, "Rowcount Sales", and display the row count from the sales table.
- Make a note of the number of records.
- Apply a data source filter on Null Invoices and keep the "false" value.
- Similarly, apply a data source filter on State Code Filter to keep the "true" value.
- Make a note of the row count value from the sales table again.
- **What is the count of sales after applying the filters? If 12946 please proceed**

Step 19 – Market Basket Analysis

- Due to the self-join on the sales table, we have two copies of all attributes.
- Let us standardize attribute names and values.
- The market basket data source has two tables. The second instance of the table is renamed as **sales-self-join**. But the fields from the second table are not renamed.
- Create a new sheet, "**Product Description Variations**".
- Create a calculated field **Product Description (self-join)** Create a similar field, **Product Description (sales)**.

Product Description (self-join) Market Basket

```
if [Description (self join)] = "Indoor Pet Camera (Wi-Fi)"  
then "Indoor Pet Camera"  
ELSE [Description (self join)] END
```

Product Description (Sales) Market Basket

```
if [Description] = "Indoor Pet Camera (Wi-Fi)"  
then "Indoor Pet Camera"  
ELSE [Description] END
```

Step 20– Market Basket Analysis

- Due to the self-join on the sales table, we have two copies of all attributes.
- Let us standardize attribute names and values.
- The market basket data source has two tables. The second instance of the table is renamed as **sales-self-join**. But the fields from the second table are not renamed.
- Create a new sheet, "**Product Description Variations**".
- Create a calculated field **Product Description (self-join)** Create a similar field, **Product Description (sales)**.

Product Description (self-join) Market Basket

```
if [Description (self join)] = "Indoor Pet Camera (Wi-Fi)"  
then "Indoor Pet Camera"  
ELSE [Description (self join)] END
```

Product Description (Sales) Market Basket

```
if [Description] = "Indoor Pet Camera (Wi-Fi)"  
then "Indoor Pet Camera"  
ELSE [Description] END
```

Step 21– Market Basket Analysis

- Switch to the old data source
- Create a new calculated field “product description and do the same as the last slide

```
if [Description] = "Indoor Pet Camera (Wi-Fi)"  
then "Indoor Pet Camera"  
ELSE [Description] END
```


Step 22– Market Basket Analysis

- Generate this table and try your best to interpret the results?

Product Description Variation

Product Description ..	Description (self join)	Stock Co..	Sales	Sales (se..
Canned Cat Food	Canned Cat Food	21137	992	1,980
ChomChom Pet Hair ..	ChomChom Pet Hair ..	22114	32,360	23,324
Dog and Puppy Pads	Dog and Puppy Pads	85123A	247,787	62,202
Earth Rated Dog Po..	Earth Rated Dog Po..	85099B	266,431	82,678
Indoor Pet Camera	Indoor Pet Camera	22502	19,861	14,177
	Indoor Pet Camera (..	22502	7,508	1,824
Litter Slide Multi-Ca..	Litter Slide Multi-Ca..	22086	8,355	16,649
Memory Foam Pet B..	Memory Foam Pet B..	47566	183,616	91,275
Milk-Bone MaroSnac..	Milk-Bone MaroSnac..	22386	27,547	17,356
NexGard Chew for D..	NexGard Chew for D..	82484	65,077	26,903
Pet Hair Remover	Pet Hair Remover	23203	108,902	34,236
Pet Odor Eliminator	Pet Odor Eliminator	84879	110,183	44,997
ProBiotic Suppleme..	ProBiotic Suppleme..	79321	140,535	36,317
Purina ONE SmartBL..	Purina ONE SmartBL..	23298	98,934	40,616
Purina Pro Plan Adul..	Purina Pro Plan Adul..	23284	56,948	31,857
Rechargeable Pet N..	Rechargeable Pet N..	22960	106,902	35,696
Sheba Perfect Porti..	Sheba Perfect Porti..	22197	52,259	20,164
Taste of the Wild Hi..	Taste of the Wild Hi..	22423	67,160	90,499
Templation Soft Cat ..	Templation Soft Cat ..	23084	89,070	36,823
UltraSound Dog Bar..	UltraSound Dog Bar..	22720	74,412	32,200

Step 23– Market Basket Analysis

- Find correlations between **Quantity** and **Quantity (self-join)**.
- Create a calculated field “**Correlation**”.
- You need to use the **INCLUDE()** and **CORR()** functions.
- Including **Invoice No** in detail ensures that the correlation is calculated at the individual checkout level.

```
CORR(  
  { INCLUDE [Invoice No]: SUM([Quantity]) },  
  { INCLUDE [Invoice No]: SUM([Quantity (self-join)]) }  
)
```

- Create a new sheet, "Market Basket Visualization", and display the correlation matrix by Stock Code and Stock Code (self-join) as a Highlight Table.
- Round Correlation to one decimal point and add Correlation and Product Description (self-join) to the table.

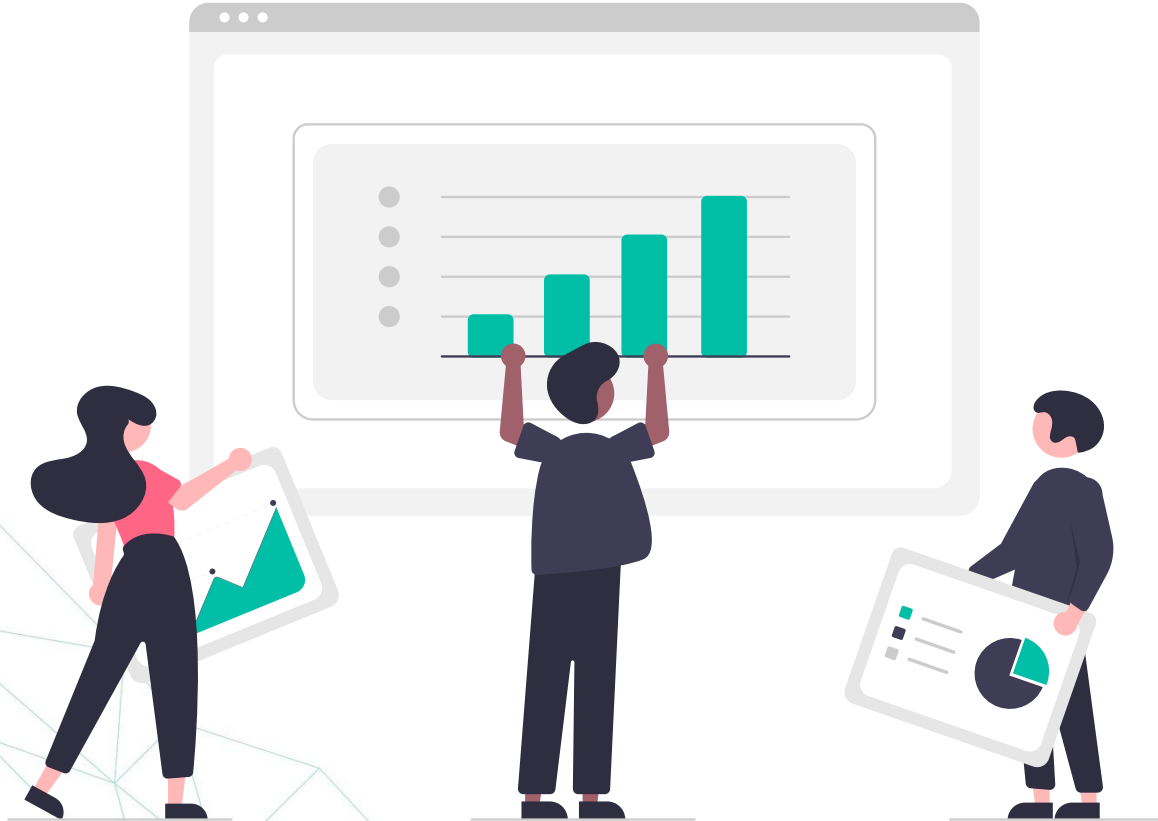
Step 24– Market Basket Analysis

- Enhance your visualization by hiding Stock Code (self-join), change colors (optional), and add a legend and a tooltip with the product description field from both sales and sales-self-join.
- Create a "Market Basket Analysis" dashboard and add the Market Basket Visualization sheet to this dashboard.
- What is the correlation between pet hair remover (23203) and indoor pet camera (22502)? If -0.1 process**

Market Basket Visualization

Product Description (self-join)	20725	21137	22086	22114	22197	22386	22423	22502	22720	22960	23084	23203	23284	23298	47566	79321	824
Canned Cat Food	0.4																
Litter Slide Multi-Cat Scented C...	0.4	0.3															
ChomChom Pet Hair Remover - ...	0.6	0.2	0.3														
Sheba Perfect Portions Pat We...	0.4	0.6	0.9	0.3													
Milk-Bone MaroSnacks Dog Tre...	0.7	0.9	0.5	0.1	0.6												
Taste of the Wild High Prairie G...	0.5	0.3	0.5	0.6	0.3	0.5											
Indoor Pet Camera	0.6		0.8	0.0	0.3	0.1	0.4										
UltraSound Dog Barking Deterr...	0.9	0.0	0.1	0.7	0.5	0.6	0.9	0.5									
Rechargeable Pet Nail Grinder	0.9	1.0	0.4	0.6	0.5	0.3	0.6	0.2	0.8								
Templation Soft Cat Treats	0.6	0.0	0.8	0.3	0.8	0.9	0.4		0.6	0.5							
Pet Hair Remover	0.8	0.0	0.9	0.9	0.8	0.7	0.8	-0.1	0.5	0.5	0.8						
Purina Pro Plan Adult Sensitive...	0.6	0.2	0.2	0.3	0.3	0.3	0.9	0.3	0.3	0.9	0.0	0.9					
Purina ONE SmartBlend Natura...	0.6	0.1	0.3	0.2	0.2	0.5	0.5	0.2	0.6	0.2	0.0	0.6	0.4				
Memory Foam Pet Beds for Sm...	0.6	0.2	0.2	0.2	0.2	0.7	0.4	0.5	0.6	0.6	0.0	0.7	0.3	0.7			
ProBiotic Supplements for Dogs	0.6	0.2	0.4	0.3	0.4	0.5	0.4	0.7	0.2	0.3	0.3	0.4	0.4	0.1	0.4		
NexGard Chew for Dogs	0.7		0.7	0.1	0.7	0.3	0.4	0.9	0.8	0.3	0.0	0.8	0.6	0.6	0.3	0.6	
Pet Odor Eliminator	0.6	0.8	0.4	0.5	1.0	0.8	0.7	0.4	0.7	0.6	0.7	0.9	0.2	0.3	0.3	0.5	
Earth Rated Dog Poop Bags	0.8	0.7	0.4	0.2	0.6	1.0	0.5	0.3	0.6	0.7	0.3	0.9	0.5	0.7	0.6	0.7	
Dog and Puppy Pads	0.7	0.1	0.5	0.2	0.6	0.7	0.3	0.4	0.4	0.5	0.2	0.7	0.4	0.5	0.5	0.6	

Let's practice



Any Questions ?

Thank You